

Dart Assignment

Module 1: Introduction to Mobile Development and Flutter

1. Benefits of using Flutter over other cross-platform frameworks

Flutter offers several advantages compared to frameworks like React Native or Xamarin:

Single Codebase

Write one code and run on Android, iOS, Web, and Desktop.

Fast Performance

Uses native compilation (no bridge like React Native), so apps are smoother.

Hot Reload

Instantly see code changes without restarting the app.

Rich UI (Widgets)

Comes with customizable widgets for attractive UI design.

Open Source & Backed by Google

Strong community and regular updates.

Consistent UI Across Platforms

Same design on Android and iOS (no dependency on native components).

2. Role of Dart in Flutter & its advantages

Dart is the programming language used in Flutter.

Role:

Used to write Flutter apps

Builds UI using widgets

Handles logic, state, and backend interaction

Advantages:

Fast Execution → Compiled to native code

Hot Reload Support → Improves development speed

Easy Syntax → Similar to Java/C#

Strong Typing → Reduces errors

Asynchronous Programming → Better for APIs and UI responsiveness

3. Steps to set up Flutter development environment

Step 1: Install Flutter SDK

Download from: <https://flutter.dev>

Extract and set PATH

Step 2: Install required tools

Install Android Studio or VS Code

Install Flutter & Dart plugins

Step 3: Setup Android Emulator

Open Android Studio → AVD Manager → Create Virtual Device

Step 4: Verify installation

flutter doctor

Step 5: Fix any issues shown

Install missing dependencies

4. Basic Flutter App Structure

main.dart

Entry point of Flutter app

Contains main code

► main() function

```
void main() {  
  runApp(MyApp());  
}
```

```
}
```

Starts the app

Calls root widget

Widget Tree

Flutter UI is made of widgets

Everything = widget (Text, Button, Layout)

Example:

MaterialApp

└─ Scaffold

└─ Center

└─ Text

```
C:\Users\popat>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.35.1, on Microsoft Windows [Version 10.0.26200.8457], locale en-IN)
[✓] Windows Version (Windows 11 or higher, 25H2, 2809)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.0.0)
[✓] Chrome - develop for the web
[✓] Visual Studio - develop Windows apps (Visual Studio Community 2022 17.14.13 (August 2025))
[!] X Visual Studio is missing necessary components. Please re-run the Visual Studio installer for the "Desktop development with C++" workload, and include
    these components:
        MSVC v142 - VS 2019 C++ x64/x86 build tools
        - If there are multiple build tool versions available, install the latest
        C++ CMake tools for Windows
        Windows 10 SDK
[✓] Android Studio (version 2025.3.4)
[✓] IntelliJ IDEA Ultimate Edition (version 2026.1)
[✓] VS Code (version unknown)
[✓] X Unable to determine VS Code version.
[✓] Connected device (3 available)
[✓] Network resources

! Doctor found issues in 1 category.
```